

FLATTRIKE · REQUIREMENTS

FlatTrike requirements

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DOCUMENT CONTROL

Document ID	FTK-REQ-001	Version	0.9
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/flattrike		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.9	2026-10-03	Amish Chadha	Status from FTK-CAL-001 v0.7 with the knuckle forks turned to trail (35 mm mechanical trail, decided by Amish 2026-10-03): wheelbase 1.415 m, turning circle 5.52 m, 71.6 kg empty, prototype \$736, production about \$471; no status change
0.8	2026-10-02	Amish Chadha	Status from FTK-CAL-001 v0.6 with the 2026-10-02 decisions carried into the model and calculations (steering lock stops, rating 148 and 138 kg, turning circle through the built linkage); no status change
0.7	2026-10-02	Amish Chadha	R1 rating restated (148 and 138 kg), R8 relaxed to 72 kg and R14 to 60 min for knuckle post plates on 2026-10-02 (FTK-DDR-003 A1 and A2); R1, R8 and R14 met on paper; R7 accepted at 4.04 h
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.5	2026-10-01	Amish Chadha	Status from FTK-CAL-001 v0.3 on the constructable design (FTK-DDR-003);

			R1, R8 and R14 not met, open decisions in FTK-DEC-001
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 status from FTK-CAL-001; R15 budget scope redefined; new R18 production cost target (FTK-DDR-001)
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status
0.1	2026-09-24	Amish Chadha	Initial scaffold

FlatTrike requirements

These requirements are checked by calculation in FTK-CAL-001 v0.7 against the constructable TRL 3 model (FTK-DDR-003). Targets are still proposals until co-design sessions with users (see FTK-PRB-001). Amish's decisions of 2026-09-25 redefine what the prototype budget covers (R15) and add a production cost target (R18) (FTK-DDR-001), and then restate R1, R8, R10 and R14 and keep R18 as it is (FTK-DDR-002). On 2026-10-02 Amish accepted the design for construction and restated R1 (148 kg of cargo with a rider up to 80 kg, 138 kg with a rider up to 90 kg), R8 (72 kg) and R14 (60 min for a knuckle post plate) (FTK-DDR-003, A1 and A2; FTK-DEC-001), so these three are met on paper. One requirement is **not met** at TRL 3: R13 without assist. R18 (production cost) is **over its value-engineering target** by USD 171. The final load label is set from the weighed trike.

The **design load case** is an 80 kg rider and 148 kg of cargo in the box on a trike of 71.6 kg: 299.6 kg gross, on a flat dry road unless stated. The upper case is a 90 kg rider with 138 kg of cargo, also 299.6 kg. FTK-CAL-001 v0.7 computes every figure at this rating.

Table 1. FlatTrike requirements and TRL 3 status (FTK-CAL-001 v0.7).

ID	Requirement	Target	Verification	TRL 3 status
R1	Carry the rated load	148 kg of cargo in the box with a rider up to 80 kg, or 138 kg with a rider up to 90 kg; gross mass 300 kg or less, within the EN 17860 cargo cycle class (restated, FTK-DDR-002; rating restated 2026-10-02, FTK-DDR-003 A1; final label set from the weighed trike)	Mass budget (FTK-CAL-001 section 3)	Met on paper: 299.6 kg in both cases at the 2026-10-02 rating (FTK-DEC-001)
R2	Frame strength	Frame survives a static proof load of twice the rated payload with no permanent set, and a fatigue load case representative of commercial use under EN 17860	Hand calculation (section 6); FEA next; proof test later	At risk: stresses 41 MPa or less and fatigue ranges below the limit, but stay buckling has a safety factor of only 2.16, the knuckle post columns reach 41 MPa and the tab-and-slot joints need FEA
R3	No welding	Every steel part is flat 3 mm plate cut from a DXF file by laser or waterjet; no bending, welding or brazing; joints bolted with locknuts	Model review	Met: 37 plates and 78 spacer washers, all flat, joined by tabs, slots and bolts; head tubes, seat tube and spacer tubes are bought sections cut to length
R4	One standard sheet	All steel frame parts nest on one 1,250 x 2,500 x 3 mm sheet (4 x 8 ft class)	Nesting check (section 2)	Met: 1,985 mm of sheet used by a true-shape nest; bounding rectangles alone would need 2,669 mm
R5	Standard bicycle parts	Wheels (20 in, ISO 406), hubs, brakes, forks (20 in), headsets (1 1/8 in), bottom bracket, cranks, chain (1/8 in), seatpost (27.2 mm) and saddle are standard sizes sold in regional markets	Parts availability survey with the partner	Met by design, with the head tubes and bottom bracket shell as bought steel sections (decided); survey pending
R6	Ships flat	Complete kit, wheels included, packs into 0.4 m ³ or less with no piece longer than 1.0 m	Packing estimate	Met: about 0.31 m ³ ; longest piece 0.85 m
R7	Assembled with hand tools	Two people assemble the kit in 4 h or less with spanners to 17 mm, hex keys, cone spanners and a screwdriver; no jig, because the plates locate each other with tabs and slots	Work-content estimate; later timed build	At risk: about 4.07 h for 97 bolt sets, the headsets and the steering linkage, about 4 minutes over; accepted 2026-10-02, to be timed at TRL 4
R8	Light enough to ride and push	Empty mass 72 kg or less including the box (relaxed from 55 kg, FTK-DDR-002, and from 70 kg on 2026-10-02, FTK-DDR-003 A1); still lighter than the 80 kg traditional rickshaw documented by ITDP	Mass budget	Met on paper: 71.6 kg with the steering lock stops and the forks turned to trail, 0.4 kg under (FTK-DEC-001)
R9	Fits market lanes	Overall width 1.0 m or less; length 2.2 m or less; turning circle 6 m or less	Model and steering geometry	Met: 0.986 m wide, 2.15 m long, 5.52 m turning circle through the built linkage at the 40° inner lock set by the lock stops (5.86 m upper bound); the trailing contact

ID	Requirement	Target	Verification	TRL 3 status
				patches shorten the wheelbase to 1.415 m
R10	Resist tipping in turns	Lateral acceleration at the tipping point 0.30 g or more loaded, and 0.22 g or more with the rider only together with a cornering-speed label (rider-only case relaxed, FTK-DDR-002)	Stability calculation (section 5); later tilt-table test	Met: 0.42 g loaded and 0.24 g rider only at any lock (lowest straight ahead; the trailing outer contact moves out as the wheels turn); label reads 6 km/h for full-lock turns when empty
R11	Stop and park safely	Stop from 15 km/h in 6 m or less at gross mass on a dry road; drum brakes on all three wheels; a parking brake that holds the loaded trike on a 10 % grade	Braking calculation (section 7); later field test	At risk: each drum needs about 37 N·m (rating not yet confirmed); fade after about 1.9 km of loaded 5 % descent
R12	Pedal on the flat	Cruise at 7 km/h or more at gross mass on a flat dirt road with 110 W at the pedals	Power calculation	Met: 7.6 km/h
R13	Climb local hills	Climb a 5 % grade for 100 m at 4 km/h or more at gross mass without the rider dismounting	Power and gearing calculation	Not met without assist: 239 W needed at the pedals. Met with assist route C (6.0 km/h), which is not in the prototype
R14	Repairable locally	Any single plate or standard part replaced in 30 min or less with hand tools, except a spine side plate in 90 min or less (FTK-DDR-002) and a knuckle post plate in 60 min or less (FTK-DDR-003 A2, 2026-10-02); open DXF files let any laser shop re-cut one plate	Bolt-count estimate; later repair trial	Met on paper: knuckle post plate about 53 min (both collar plates and the headset cups come off), within 60 min; spine side plate about 80 min, within 90 min
R15	Affordable prototype	Parts for the pedal-only prototype cost at or below the value-engineering target of \$800 (<code>project.yaml</code> <code>budget_usd</code> , a hypothetical control target). The optional assist kit and any SwapCell pack are outside this budget; a SwapCell pack is priced once in the SwapCell repo	Priced BOM (<code>bom/</code> <code>bom.csv</code>)	Within the value-engineering target: \$736 against \$800 (USD 64 under)
R16	Last in commercial service	Frame life of 5 years or more in daily outdoor commercial use, against 2 to 3 years for traditional wooden rickshaw frames (ITDP)	Corrosion and fatigue review; later field trial	Not verifiable at TRL 3: unprotected steel could lose up to 0.42 mm in 5 years; depends on coating and sealed joints
R17	Work as a stall	Lockable box of 150 L or more; box lid usable as a counter at 0.75 to 0.95 m height	Model check	Met: 151 L inside; lid at 0.84 m
R18	Competitive in production	Production cost at or below the value-engineering target of \$300 per trike at 100 units (decided by Amish, 2026-09-25; kept, FTK-DDR-002)	Cost estimate (section 9); later partner quotes	Over the value-engineering target by USD 171: about \$471, over two thirds of it bought bicycle parts; wholesale prices wait on a partner

■ Assumptions

- Rolling resistance coefficient about 0.015 on a mix of worn asphalt and packed dirt; drag area about 0.9 m² for an upright rider behind a cargo box; air density 1.2 kg/m³.
- The rider sustains about 110 W at the pedals for an hour; short efforts can reach 200 W or more.
- Drivetrain efficiency (chain and 3-speed hub) about 89 %.
- Plate is 3 mm hot-rolled mild steel (S235JR or ASTM A36 class), density 7,850 kg/m³.
- Plywood is 9 to 12 mm exterior grade, about 700 kg/m³.
- Volume prices (R18) assume one full sheet at \$1.0/kg, cutting at \$0.30/m and bought parts at 65 % of prototype retail; no quotes yet.
- The 300 kg gross limit follows the EN 17860 class for single and multi-track cargo cycles; the standard's detailed test loads have not yet been checked against this design.